

EDUCATION

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|---|---|
| <ul style="list-style-type: none"> Purdue University Ph.D. in Computer Science <ul style="list-style-type: none"> Advisor: Dr. Ruqi Zhang | IN, USA
<i>Aug. 2025 – 2030 (Expected)</i> |
| <ul style="list-style-type: none"> Tianjin University B.S. in Data Science and Big Data Technology, School of Mathematics <ul style="list-style-type: none"> Advisor: Dr. Bing Cao & Dr. Qinghua Hu | Tianjin, China
<i>Aug. 2021 – Jun. 2025</i> |

RESEARCH INTERESTS

My research focuses on developing reliable and efficient post-training algorithms for foundation models, with an emphasis on improving the general intelligence and safety awareness of LLMs. I am particularly interested in how models can learn and self-evolve from limited, noisy, or imperfect supervision.

- LLM/VLM Post-Training**
 - Reinforcement Learning, On-Policy Distillation, Reasoning, Self-Correction, and Self-Evolving*
- Alignment and Reliability**
 - Preference Alignment, Safety Alignment, Reliable Supervision, and Robustness*

PUBLICATIONS

(* denotes equal contribution)

- [P1] **Addressing Performance Saturation for LLM RL via Precise Entropy Curve Control**
 Bolian Li, Yifan Wang, **Yi Ding**, Anamika Lochab, Ananth Grama, Ruqi Zhang
Preprint  
- [P2] **SafeWork-R1: Coevolving Safety and Intelligence under the AI-45° Law**
 Shanghai AI Lab, ..., **Yi Ding**, [100+ Authors]
Technical Report 
- [C1] **Learning Self-Correction in Vision–Language Models via Rollout Augmentation**
Yi Ding, Ziliang Qiu, Bolian Li, Ruqi Zhang
International Conference on Machine Learning (ICML 2026)  
- [C2] **Position: Modular Safety Guardrails Are Necessary for Foundation-Model-Enabled Robots in the Real World**
 Joonkyung Kim, Wenxi Chen, Davood Soleymanzadeh, **Yi Ding**, Xiangbo Gao, Zhengzhong Tu, Ruqi Zhang, Fan Fei, Sushant Veer, Yiwei Lyu, Minghui Zheng, Yan Gu
International Conference on Machine Learning (ICML 2026) 
- [C3] **Rethinking Bottlenecks in Safety Fine-Tuning of Vision Language Models**
Yi Ding*, Lijun Li*, Bing Cao, Jing Shao
International Conference on Learning Representations (ICLR 2026)  
- [C4] **Sherlock: Self-Correcting Reasoning in Vision-Language Models**
Yi Ding, Ruqi Zhang
Neural Information Processing Systems (NeurIPS 2025)  
- [C5] **Visual Contextual Attack: Jailbreaking MLLMs with Image-Driven Context Injection**
 Ziqi Miao*, **Yi Ding***, Lijun Li, Jing Shao
Empirical Methods in Natural Language Processing Main Conference (EMNLP 2025) 
- [C6] **ETA: Evaluating Then Aligning Safety of Vision Language Models at Inference Time**
Yi Ding, Bolian li, Ruqi Zhang
International Conference on Learning Representations (ICLR 2025)  

[C7] **Test-Time Dynamic Image Fusion**

Bing Cao (Advisor), Yinan Xia*, **Yi Ding***, Changqing Zhang, Qinghua Hu
Neural Information Processing Systems (NeurIPS 2024) 

[C8] **Predictive Dynamic Fusion**

Bing Cao (Advisor), Yinan Xia*, **Yi Ding***, Changqing Zhang, Qinghua Hu
International Conference on Machine Learning (ICML 2024) 

AWARDS

- **NSF ACCESS Discover Project Award** 2025-2026
- LLM Post-Training

RESEARCH EXPERIENCE

- **RZ-Lab, Purdue University** **IN, USA**
▪ *Research Assistant*, Advised by **Dr. Ruqi Zhang** *May 2024–May 2025*
- **Open Trust Lab, Shanghai AI Laboratory** **Beijing, China**
▪ *Research Scientist*, Advised by **Dr. Lijun Li** *Dec. 2024–Mar. 2025*
- **MLDM Lab, Tianjin University** **Tianjin, China**
▪ *Research Assistant*, Advised by **Dr. Bing Cao** and **Dr. Qinghua Hu** *Sep. 2023–Dec. 2024*

SKILL

Languages: Chinese Mandarin (Native), English (TOEFL 102(22))

Research Abilities: Proficient in coding: Python, $\LaTeX$, MATLAB; Enjoys mathematical derivations; Solid foundation in mathematics and statistics.

SERVICE

- **Conference Reviewer**
- ICML 2026; NeurIPS 2025,2026; ICLR 2025,2026; ARR 2025 May